

D A Y 1 0

Basics of Cloud Computing

Lesson 1: AWS (Amazon Web Services)

Digital Science Program | Graduate Level

Cloud Computing & Big Data Analytics

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Today's Agenda

1. What is AWS and why should you care?
2. The origin story — from bookstore to cloud giant
3. Core AWS services in plain language
4. Global infrastructure — where your data lives
5. Pricing — the pay-as-you-go revolution
6. AWS for Data Science — the toolkit
7. Security fundamentals — IAM & Shared Responsibility
8. Hands-on: your first steps with AWS

What Is AWS?

Amazon Web Services — the world's leading cloud computing platform

AWS (Amazon Web Services) — Defined

A cloud computing platform offering 200+ on-demand services over the internet, billed by actual usage — no upfront hardware purchase.

200+

Services

34

Regions

31-33%

Market Share

<2 min

Server Launch

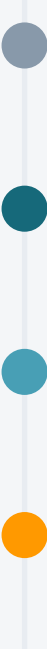
- Compute, storage, databases, ML, networking, IoT, security
- ~31-33% of global IaaS market (Gartner, Synergy Research Group)
- Larger market share than Microsoft Azure + Google Cloud Platform combined (IaaS)

Why Data Scientists Need AWS

Factor	Without Cloud	With AWS
Upfront Cost	\$4,000 GPU workstation	\$0
Per-Hour Cost	\$0 (already paid)	~\$0.53/hr
20-hr Training	\$4,000	\$10.52
Setup Time	Days to weeks	~2 minutes
Idle Cost	Depreciating asset	\$0 (shut it down)
Scaling	Buy another machine	Click a button

Key Takeaway: Cloud converts CapEx (Capital Expenditure) into OpEx (Operational Expenditure) — pay only for what you use.

The AWS Origin Story

- 
- 2000s** Amazon battles holiday traffic spikes — servers idle 90% of the year
 - 2006** AWS launches with 3 services: S3, SQS, EC2
 - 2010s** Rapid expansion — databases, ML, IoT, analytics services added
 - 2025** 200+ services, 34 Regions, 108+ AZs across every continent

Key Insight: AWS was born from Amazon's own scaling problem — the same problem every data project faces.

Core AWS Services

200+ services — but ~15 cover 90% of what you'll use in data science

Core Services — Compute (The Engines)



EC2 (Elastic Compute Cloud)

Virtual servers by the second — pick CPU, RAM, GPU, OS. Like renting a furnished apartment.



Lambda (Serverless Functions)

Run code without servers — pay per millisecond. Like hiring a freelancer for one task.



ECS / EKS (Container Services)

Run Docker containers at scale with managed orchestration. Portable, reproducible.

Key Takeaway: EC2 = full control over a virtual machine. Lambda = zero server management. Choose based on your workload.

Core Services — Storage (The Filing Cabinets)

S3

S3 (Simple Storage Service)

Object storage for any file. 99.999999999% durability (eleven 9s). Infinite capacity.

EB

EBS (Elastic Block Store)

SSD-backed drives that attach to EC2 instances. Like virtual hard disks.

GL

Glacier (Archival Storage)

Ultra-cheap storage for rarely accessed data. Retrieval takes minutes to hours.

Key Takeaway: S3 is the backbone — nearly every AWS workflow starts and ends with S3.

Core Services — Database (The Organized Data)



RDS (Relational Database Service)

Managed MySQL, PostgreSQL, Oracle. AWS handles backups, patches, scaling.



DynamoDB (NoSQL Key-Value)

Single-digit millisecond latency. Automatic scaling. No server management.



Redshift (Data Warehouse)

Columnar analytics warehouse. Complex SQL queries across billions of rows.

Core Services — Machine Learning (The Brains)



SageMaker (Full ML Platform)

Build, train, deploy ML models end-to-end. Studio IDE, distributed training, one-click deploy.



Comprehend (NLP Service)

Extract sentiment, entities, topics from text. Pre-trained deep learning models.



Rekognition (Computer Vision)

Analyze images and video for objects, faces, text, scenes. API-based.

Key Takeaway: SageMaker is your primary tool — covers the entire ML lifecycle from notebook to production endpoint.

Core Services — Networking & Security

Networking (The Roads)

- VPC (Virtual Private Cloud) — Your isolated network
- Route 53 (DNS) — Domain name resolution
- CloudFront (CDN) — 400+ edge locations worldwide

Security (The Guards)

- IAM — Who can access what
- KMS (Key Management) — Encryption keys
- CloudTrail — Audit log of every API call

Real-World Data Pipeline on AWS

Pipeline Step	AWS Service	What It Does
Ingest	Kinesis	Stream real-time review data
Clean	Lambda	Trigger on S3 upload, clean text
Store	S3	Raw + cleaned data files
Analyze	Comprehend / SageMaker	Sentiment classification
Query	Redshift / Athena	Analytical SQL queries
Visualize	QuickSight	Interactive dashboards
Secure	VPC + IAM	Network isolation + access control

AWS Global Infrastructure

34 Regions · 108+ Availability Zones · 400+ Edge Locations

Global Infrastructure – Three-Level Hierarchy

1

Regions (34)

Geographically isolated areas (e.g., us-east-1 = N. Virginia). You choose at deployment. Affects latency, compliance, cost.

2

Availability Zones (108+)

Separate data centers within a Region, connected by high-speed fiber. 3+ AZs per Region for fault tolerance.

3

Edge Locations (400+)

Caching facilities for CloudFront CDN + Route 53 DNS. Bring content closer to end users.

Why Region Selection Matters

Factor	Impact
Latency	Montreal → ca-central-1: ~3ms vs. ap-southeast-1: ~210ms
Compliance	GDPR requires EU citizen data stored in EU regions
Cost	Prices vary — us-east-1 (Virginia) often cheapest
Services	Not all services available in all regions

Key Takeaway: "The cloud" has a physical location — choose wisely based on users, laws, and budget.

AWS Pricing

The pay-as-you-go revolution — four models to fit every budget

Four Pricing Models

Model	Commitment	Discount	Best For
On-Demand	None	0%	Unpredictable workloads
Reserved (RIs)	1–3 years	30–72%	Steady-state servers
Spot Instances	None (reclaimable)	Up to 90%	ML training, batch jobs
Savings Plans	\$/hr commitment	Up to 72%	Flexible compute

Spot Instances are perfect for ML training!

Same hardware, 60-90% cheaper. AWS can reclaim with 2-min warning → use checkpointing.

The AWS Free Tier — Your Starting Point

EC2

750 hrs/month t2.micro or t3.micro (runs 24/7 for a year)

S3

5 GB standard storage

DynamoDB

25 GB storage

Lambda

1 million invocations/month

SageMaker

250 hours/month Studio notebook (first 2 months)

Key Takeaway: Always estimate costs BEFORE deploying — use the AWS Pricing Calculator (calculator.aws)

Cost Example — Monthly Thesis Project

Resource	On-Demand	Spot + Free Tier
50 GB S3 storage	\$1.15	\$1.15
80 hrs GPU training	\$244.80	~\$73.60
24/7 API server	\$7.59	\$0.00
TOTAL	~\$254	~\$75

SAVINGS: 70%+

\$254 → \$75/month with smart pricing choices

AWS for Data Science

The complete toolkit — from data ingestion to model deployment

The Data Science Toolkit on AWS

Stage	Service	Purpose
Storage	S3	Raw & processed datasets
Data Prep	AWS Glue	Serverless ETL
Ad-hoc Query	Athena	SQL directly on S3 data
Big Data	EMR	Managed Hadoop/Spark
Streaming	Kinesis	Real-time ingestion
ML Platform	SageMaker	Train, tune, deploy models
BI / Dashboards	QuickSight	Interactive visualizations

Amazon SageMaker — Your ML Command Center

SageMaker Studio

Cloud IDE with Jupyter notebooks, experiment tracking, model monitoring

SageMaker Training

Distributed training across dozens of GPU instances in parallel

SageMaker Endpoints

One-click HTTPS deployment for real-time predictions

SageMaker Pipelines

CI/CD for ML — reproducible, automated workflows

AWS Security Fundamentals

IAM (Identity & Access Management) and the Shared Responsibility Model

The Shared Responsibility Model

AWS Secures "OF the Cloud"

- Physical data centers & hardware
- Networking infrastructure
- Hypervisor layer
- Managed service infrastructure
- Fire suppression & physical access

YOU Secure "IN the Cloud"

- Your data & encryption choices
- IAM configuration (users, roles)
- Network settings (VPC, security groups)
- Application code & dependencies
- Operating system patches

Analogy: AWS builds & secures the apartment building. You secure your own apartment — who has keys, what's inside.

IAM — Four Building Blocks



Users

Individual accounts with unique credentials



Groups

Collections sharing permissions (e.g., "DataScientists")



Roles

Temporary permissions for services/apps to assume



Policies

JSON documents defining allowed/denied actions on resources

Golden Rule: Principle of Least Privilege — every entity gets ONLY the minimum permissions needed.

Least Privilege in Practice – Team Example

Team Member	Role	CAN Access	CANNOT Access
Alice	ML Engineer	SageMaker, S3, EC2	Billing, IAM Admin
Bob	Data Engineer	Glue, S3, Redshift, EMR	SageMaker, Billing
Carol	Team Lead	All + Billing (read)	IAM Admin
Dave	Intern	S3 (read), Studio (read)	Everything else

Misconfigured IAM is the #1 cause of cloud security incidents. Get this right first.

Your First Steps with AWS

Five actions you can take today — right now

Five Actions — Start Today

1

Create Free Account

aws.amazon.com/free — requires credit card, email, phone

2

Secure Immediately

Enable MFA on root, create IAM user, set \$5 billing alert

3

Launch First EC2

Amazon Linux 2023 → t3.micro (Free Tier eligible)

4

Create First S3 Bucket

Globally unique name → upload a file

5

Open SageMaker Studio

Launch Jupyter notebook — start experimenting with ML

Two Ways to Interact with AWS

AWS Management Console

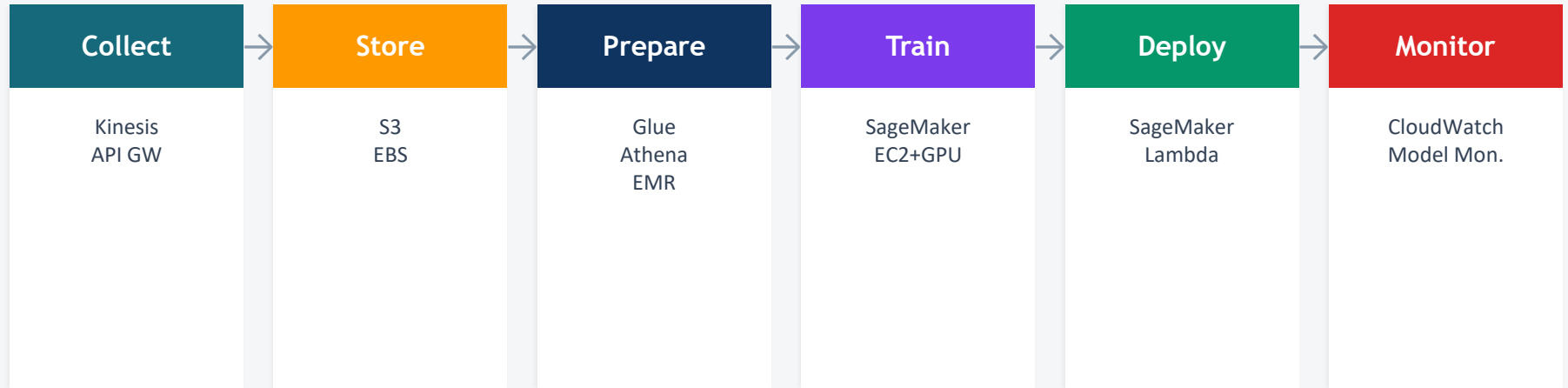
- Visual, point-and-click (browser)
- Great for learning & exploration
- `console.aws.amazon.com`

AWS CLI (Command Line)

- Terminal-based commands
- Essential for automation & scripting
- `pip install awscli`

Also: AWS SDKs (Python boto3, Java, Node.js), CloudFormation (IaC), CDK

End-to-End Data Science Workflow on AWS



S3 is the central hub connecting all stages | IAM controls access at every step | VPC isolates the pipeline

Recommended Learning Resources



YouTube Videos

Search: "AWS In 10 Minutes | Simplilearn" or "AWS Tutorial 2025 | Kevin Stratvert"



AWS Skill Builder

Free digital training — skillbuilder.aws



Well-Architected Framework

Architecture best practices — aws.amazon.com/architecture/well-architected/



AWS Pricing Calculator

Estimate costs before deploying — calculator.aws

Note: Videos are the IP of their respective creators. No affiliation claimed. Verify links with [virustotal.com](https://www.virustotal.com)

Key Abbreviations Reference (1/2)

Abbr.	Full Name	Abbr.	Full Name
AWS	Amazon Web Services	ML	Machine Learning
EC2	Elastic Compute Cloud	NLP	Natural Language Processing
S3	Simple Storage Service	GPU	Graphics Processing Unit
RDS	Relational Database Service	CDN	Content Delivery Network
IAM	Identity & Access Management	DNS	Domain Name System
VPC	Virtual Private Cloud	ETL	Extract, Transform, Load
IaaS	Infrastructure as a Service	API	Application Programming Interface
AZ	Availability Zone	MFA	Multi-Factor Authentication
CapEx	Capital Expenditure	OpEx	Operational Expenditure

Seven Things to Remember from Today

- 1 AWS = pay-as-you-go cloud with 200+ services — no upfront hardware
- 2 Core: EC2 (compute), S3 (storage), RDS (database), SageMaker (ML)
- 3 34 Regions, 108+ AZs — region choice affects latency, compliance, cost
- 4 Pricing: On-Demand, Reserved, Spot (up to 90% off), Free Tier
- 5 Shared Responsibility: AWS secures infrastructure, you secure your config
- 6 IAM + Least Privilege = your #1 security priority
- 7 Start today: Free account → secure it → launch EC2 → create S3 bucket

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Thank You

Questions? Let's discuss!

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